Shloke Binani

33A Canal Circular Road, Swarnamani, Kolkata-700054

+91 7003855628 | shlokebinani@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

VELLORE INSTITUTE OF TECHNOLOGY, Vellore, India 2023-2027
Bachelor of Technology in Electronics and Communication Engineering
CGPA: 8.68/10.0

TECHNICAL SKILLS

- **Languages & Core Tech**: Java (DSA), Python, C++, SQL, C, Object-Oriented Design (OOD)
- **Core Foundations**: Data Structures, Algorithms & Complexity Analysis, DBMS, Operating Systems (OS), Cloud Computing, Computer Networks (CN)
- **Backend & Cloud**: MYSQL, FastAPI, GraphQL, PostgreSQL, Neon, Docker, Vercel, Render, REST APIs
- **AI, ML & DevOps Tools**: Machine Learning & Deep Learning (QCNN, MLP, SVM, Random Forest), Data Mining, Prompt Engineering, AI Pipeline Integration, Git, Postman, CI/CD, n8n
- **Soft Skills**: Cross-Functional Collaboration, Analytical Problem-Solving, Adaptability, Techno-Functional Alignment

EXPERIENCE

Kofuku Technologies Pvt. Ltd., On-Site May 2025 - July 2025
Software Engineer Intern - Engineered backend systems to drive business efficiency and automate cross-team operations

- Improved reliability of a distributed backend system by **30%** (reduced API failure rate) by designing scalable **FastAPI** endpoints and resolving **JWT authorization** faults across services using **Postman** and **Hasura GraphQL**
- Optimized cross-team operations, eliminating **8 to 12 hours per week** of manual coordination, by architecting automated event-driven workflows and data integration pipelines in **n8n**
- Accelerated reservation speed, cutting facility booking latency by **60%**, by deploying a voice-AI agent integrating **speech-to-text APIs** and real-time calendar synchronization
- Operating within an **Agile** environment, improved software release stability and reduced QA rework cycles by **20%** by standardizing API specifications and establishing automated unit testing suites

ACADEMIC PROJECTS

ResiliNet: Decentralized 6G Network Simulator | *Python, Network Routing, Simulation* GitHub

- Designed a priority-aware bandwidth reallocation protocol for **distributed systems** in 6G networks, utilizing a **Web3 DePIN** smart contract mechanism to autonomously compensate nodes that yield non-essential bandwidth
- Simulated node-level resource contention across a **3-node** mesh testbed, applying a DPI-based **Social Urgency Taxonomy (SUT)** to reduce peak latency by **89.8%** and drop packet loss to zero under heavy load

Paradox - Scalable Ticket Booking System | *PostgreSQL, Neon, Render, Vercel* Live Link

- Engineered a stateless backend designed to handle high-concurrency traffic at **scale**, implementing strict **row-level database locking** on Neon PostgreSQL to serialize simultaneous checkout requests and prevent double-booking race conditions
- Implemented automated waitlist queues utilizing **ACID-compliant** transactions for atomic, time-limited seat reallocation upon cancellation, deploying the fault-tolerant backend via **Render** and frontend via **Vercel**

MS-HQNet: Multi-Stream Hybrid Quantum Framework | *Python, QCNN, SM-SSA, NCA*

- Designed a Hybrid **QCNN** combining a **6-qubit** quantum circuit with a classical MLP bypass for EMG signal classification, achieving **99.17% accuracy** across **600 signals**
- Constructed robust **data mining** pipelines utilizing **SM-SSA** and **NCA** to reduce algorithmic complexity and optimize feature extraction, ensuring system reliability

LEADERSHIP

Society of Automotive Engineers (SAE-VIT) | *Marketing, Design & PR Head* May 2024 - May 2026

- Directed multi-channel branding campaigns reaching **1,000+ students**, securing university administration buy-in and expanding chapter outreach
- Managed operations and **40+ volunteers** as POC for SAE AirShow 25, increasing attendee participation by **30 to 45%**
- Streamlined cross-functional logistics and team workflows across **12 major university events** during graVITas 25, ensuring 100% on-time execution

ACADEMIC ACHIEVEMENTS & CERTIFICATIONS

- Solved **250+ problems** across LeetCode & GeeksforGeeks (Easy to Hard), mastering Advanced Data Structures and SQL
- Awarded the **Elite Category Chapter Award** for SAE-VIT for excellence in operations, 2025 to 2026
- Awarded the **Leadership Certificate** for exceptional contribution and management by Vellore Institute of Technology
- Won **Runner-Up** position at Central Hack for leveraging Web3 Crisis Protocols
- Won **Runner-Up** position at Providence Leap Ideathon for AI-Driven Healthcare Benefit Orchestration

Certifications: Statistics OnRamp (2024) & MATLAB OnRamp (2024), The MathWorks, Inc.